# Vishwakarma Awards 2026–27

Stage 1 Proposal

## 1. Title of the Project / Product

Clear-Flow: Smart Manhole Monitoring and Overflow Prediction System

## 2. Problem Statement

Urban drainage today is entirely reactive — manholes only get cleared once they've already overflowed, causing traffic jams, bad smells, and a real disease risk on the street above. What's missing is a predictive system that can catch a blockage early, while there's still time to act, instead of after the street is already flooded.

## 3. Solution Proposed

Clear-Flow is a small, waterproof device that fits under an existing manhole cover and keeps track of the water level inside, without anyone having to open the cover and check by hand.

Inside the device, an ultrasonic sensor points down the shaft and works out how far the water is from the top by timing a sound echo. This reading is cleaned up locally on a small chip (ESP32-S3) built into the device, which filters out noise from splashing so only the real rising or falling trend is kept.

The whole node sits inside a sealed, waterproof enclosure, so even if the manhole fills up completely and the unit ends up underwater during a bad overflow, the electronics inside stay dry and keep working afterwards.

The device itself doesn't check the internet for anything — it only talks to a LoRaWAN gateway, so there's no SIM card, no Wi-Fi, and no mobile data involved down in the manhole. The weather check happens on the dashboard side instead: the cloud platform cross-references live weather data for that area with the incoming sensor readings, so it can tell a genuine blockage apart from water rising simply because of heavy rain outside. When it's really just rain, the dashboard raises it as a flash flood alert instead of a blockage alert, so the team knows which kind of problem they're actually walking into.

Rather than sending data all the time, the device stays in a low-power sleep mode and only wakes up to send a quick health update at regular intervals, over the long-range LoRaWAN link. If the water level starts climbing quickly, or the dashboard flags heavy rain in that area, the device breaks out of its normal cycle and switches to continuous monitoring, sending fresh readings until the situation settles back down.

The dashboard doesn't just wait for something to go wrong. It also shows live water levels for every manhole in the network on one screen, with a colour-coded heat map marking which ones are closest to a blockage or overflow at any given time. On top of this, an AI model studies how sludge and blockage levels build up in each manhole over weeks and months, and uses that pattern to automatically generate a cleaning schedule for

the maintenance team. This means manholes that are close to a blockage get cleaned before they become a problem, instead of the team waiting for a public complaint or a flood to find out.

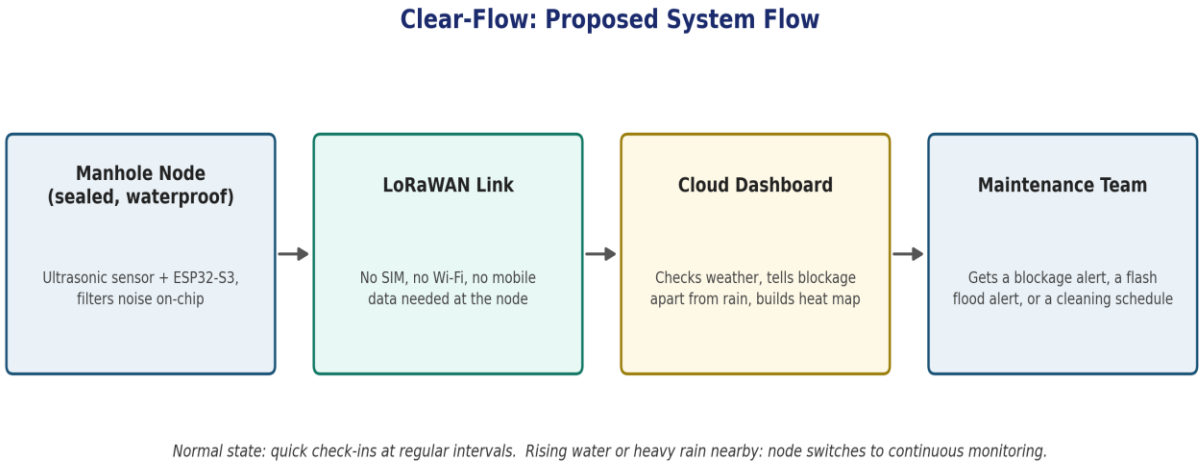



*Fig. Proposed system flow, from the sensor to the maintenance team.*

## 4. Implementation Plan

We plan to start by installing a small number of Clear-Flow units on manholes in one area, working with a local municipal body or public works department. Over a few weeks, the devices will track water levels and send this data to the dashboard, while we check how accurate the readings and alerts are against what is actually happening on the ground.

Once the readings are reliable, we will hand the dashboard over to the local maintenance team so they can start using the AI-generated cleaning schedule in place of their usual manual rounds. The main beneficiaries are municipal corporations and their sanitation staff, and indirectly, everyone who uses the roads above these drains. The expected outcome is fewer sudden floods, and the deliverable at the end of this stage is a small, working pilot network of manholes that a city can point to before deciding to scale up further.

## 5. Safety and Efficiency Analysis

Clear-Flow runs entirely on low-voltage battery power (3.7V to 5V), so there is no risk of an electrical spark inside a manhole, which can sometimes contain flammable gases. The sensor itself never touches the water below, since it measures depth using sound from above, so there is no direct contact with sewage or waste. The whole node also sits inside a sealed, waterproof enclosure, so a bad overflow that submerges the device doesn't damage the electronics or take it offline right when it's needed most.

The biggest safety benefit is indirect: since the device tells the team exactly which manholes actually need attention, workers no longer have to routinely open and enter manholes just to check if everything is fine.

Manholes can contain low oxygen levels or toxic gases, so cutting down on unnecessary entries directly reduces risk to sanitation workers.

On the efficiency side, the AI-based cleaning schedule means maintenance teams stop spending time checking manholes that do not need it, and instead focus only on the ones that are actually building up a blockage. This should reduce both emergency callouts and the total number of manhole inspections a team has to carry out in a month. Keeping the weather check on the dashboard side also keeps the node itself simple — it doesn't need any extra hardware, data plan, or battery drain just to talk to the internet.

## 6. Scalability and Future Development

Since Clear-Flow runs on LoRaWAN, a single gateway can support more than a thousand devices at once, so scaling from a handful of manholes to an entire city zone mainly means adding more gateways, not rebuilding the network from scratch.

Looking further ahead, we want to add a small gas sensor to the same device, to warn workers about dangerous gases such as hydrogen sulphide or methane before they open a manhole cover. We are also exploring whether the same ultrasonic approach can be adapted to scan horizontal pipelines, to find out exactly where along a pipe a blockage is sitting, rather than only knowing which manhole is affected. A longer-term idea is adding a small kinetic energy harvester, so that vibrations from passing vehicles slowly top up the device's battery over time.

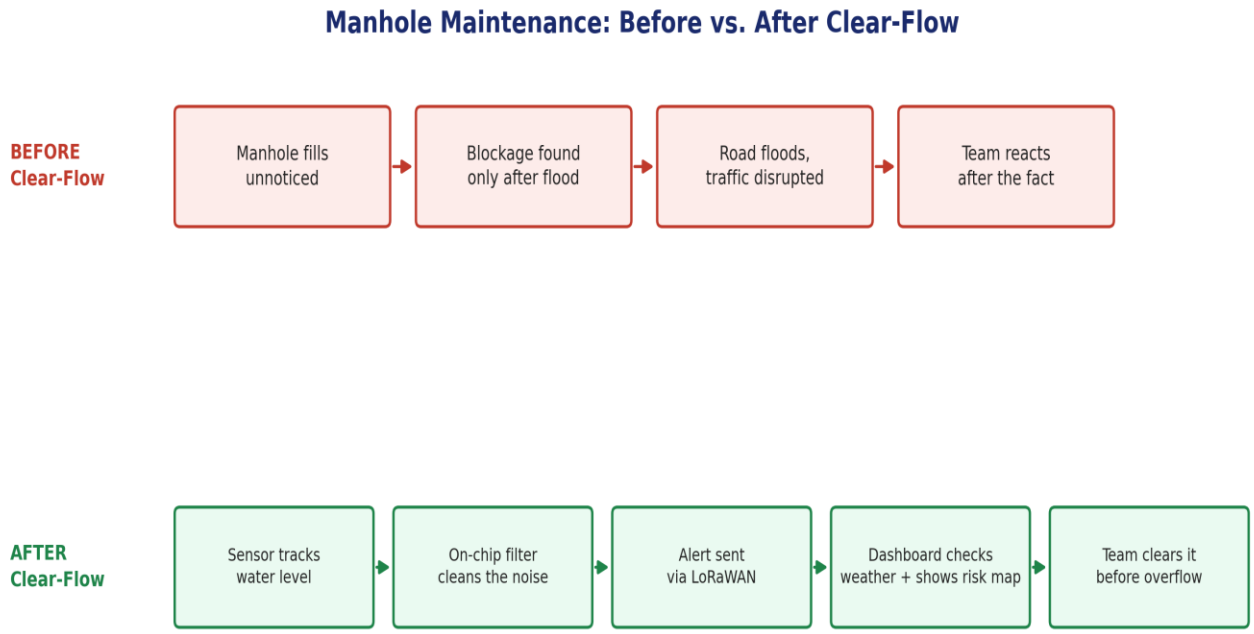
## 7. Innovation Advantage

Most sensors already sold for tank or drain monitoring are industrial products that cost lakhs of rupees per unit, which makes it unrealistic for a city to fit them on every manhole. On the other end, basic student IoT projects usually just send raw sensor data to the cloud over mobile data, which drains the battery quickly and does not work well underground.

Clear-Flow sits in between the two: it filters data on the device itself, runs on a long-range radio that needs no mobile data or SIM card, and pairs this with an AI layer that does not just display a number but actually predicts when a manhole will need cleaning. It also separates a genuine blockage from a rain-driven rise on the dashboard side, and shows every manhole's live level and risk on one map, instead of just sending alerts one manhole at a time. As far as we have found, no low-cost device currently combines on-device filtering, cloud-side weather checks, and AI-based predictive cleaning in one product built specifically for manholes.

## 8. Visual Impact & User Journey

The diagram below shows how a manhole is typically handled today, compared to how it will be handled once Clear-Flow is in place.




In short, the current process only reacts once a manhole has already overflowed, while Clear-Flow shifts the entire process earlier, so the maintenance team steps in before the road ever floods.